# Q 1

| Question | Answer | Marks |
|---|---|---|
| 8(a) | Any **one** from:<br><br>• It has electrical components. // by example<br>• It is programmable.<br>• It can move. | **1** |
| 8(b) | Any **one** from:<br><br>• Permanent **instructions**/**software** that are programmed into/stored in the ROM.<br>• **Instructions**/**software** that allow hardware to be controlled/managed.<br>• **Instructions**/**software** that provides the operating system with a platform to run on. | **1** |
| 8(c) | Any **two** from:<br><br>Examples:<br><br>• user input // by example<br>• encountering an obstacle<br>• emergency stop<br>• hardware error<br>• software error<br>• battery low<br>• component overheating<br>• two processes accessing the same location | **2** |
| 8(d) | Any **two** from:<br><br>Examples:<br><br>• Workers don't need to lift **heavy** machinery<br>• The workers are less likely to injure themselves<br>• The workers don't need to be in any dangerous environments in the factory<br>• The workers don't need to perform boring/repetitive tasks such as moving car parts<br>• Workers can focus on having higher expertise skills<br>• Create new jobs for workers to maintain the robots. | **2** |

| Question | Answer | Marks |
|---|---|---|
| 8(e) | Machine learning | **1** |

# Q 2

| Question | Answer | Marks |
|---|---|---|
| 5(a) | **Three** from:<br><br>• The <u>inference</u> engine is used …<br>• … to **decide** which questions to ask the user …<br>• … based on the **previous** data input.<br>• Symptoms input are located in/compared to knowledge base.<br>• … then applies the rule base to the knowledge base (to decide the diagnosis) | **3** |
| 5(b)(i) | Any **two** from:<br><br>Examples:<br><br>• Sensors<br>• Microprocessors<br>• Actuators | **2** |

| Question | Answer | Marks |
|---|---|---|
| 5(b)(ii) | Any **four** from:<br><br>Examples:<br><br>• a doctor doesn't need to **travel** to the hospital to do the surgery.<br>• … so it can be done by any specialist/doctor in the world.<br>• … so it can be done immediately without needing to wait for travel time // reduces the waiting time for the patient.<br>• … so the doctor may be better at the surgery as they won't be tired from travel.<br>• … so travel costs are saved.<br>• Surgery with robots enhances precision/accuracy.<br>• … so a smaller incision can be made.<br>• … so the recovery time may be shorter.<br>• … as the components used to enter the body can be much smaller than a human hand.<br>• … so the surgery is safer/ more hygienic // by example e.g. can stop the doctor needing to be near an infectious patient.<br>• The surgery may have a higher rate of success | **4** |
| 5(b)(iii) | **Two** from (one for a point and one for the matching expansion):<br><br>Examples:<br><br>• The internet connection could be lost/delayed …<br>• … so the surgery may not be able to continue.<br><br>• The robot will be expensive to buy/maintain …<br>• … this money could have been spent on other causes.<br><br>• The robot could be hacked ….<br>• … and endanger the patient's life.<br><br>• Data could be corrupted in transmission …<br>• … changing the nature of the instruction for the robot.<br><br>• The robot's hardware could malfunction …<br>• … so the surgery cannot continue | **2** |

# Q 3

| Question | Answer | Marks |
|---|---|---|
| 7(a) | • It does not have a **mechanical** structure<br>• It does not have any actuators<br>• … so it cannot move itself | **3** |
| 7(b) | Microphone | **1** |
| 7(c) | Any **four** from:<br><br>For example:<br><br>• Gather data from the same/different voices // It can ask a user to say lots of different words<br>• **Analyse/identify patterns** in a user's voice<br>• It can **adapt** its own **processes**<br>• It can learn to **identify** a **user's** voice<br>• Stores successful/unsuccessful voice commands<br>• It can learn different ways of making the same request<br>• It can learn the different pronunciations // It can learn different accents<br>• It can learn by listening to lots of different voices<br>• It can learn to ignore any background/environmental noise if a user is speaking | **4** |

# Q 4

| Question | Answer | Marks |
|---|---|---|
| 6(a) | Any **three** from:<br><br>• Simulation of intelligent behaviours by computers<br>• Collection of data and rules for using that data<br>• A computer that has the ability to learn<br>• A computer that has the ability to adapt<br>• A computer that has the ability to reason/make decisions | **3** |
| 6(b) | Any **three** from<br><br>For example:<br><br>• It could use machine learning<br>• It can collect data about the **players** actions<br>• **Analyse/identify patterns** in the player's movements<br>• … so it can learn how the player moves<br>• … to help predict the next movement of the player<br>• It can adapt its own movements to match the players // It can adapt its own processes<br>• It can store successful/unsuccessful moves<br>• It will learn the most efficient/optimal movements against a player | **3** |